

PROJECT 1: DATA CLEANING & PREPARATION

E-Commerce Sales Order Dataset

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1. Objective

To inspect the raw e-commerce sales order dataset for data quality issues, duplicates, missing values, inconsistent data types, and calculation errors and resolve any issues found before the dataset is used for downstream analysis.

2. Dataset Overview

Tool Used: Microsoft Excel

Total Records: 1,200 orders

Columns: OrderID, Date, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemsInCart, CouponCode, ReferralSource, TotalPrice

3. Cleaning Checks Performed

Check	Method	Result
Duplicate records	Checked OrderID and full-row duplicates	No duplicates found
Missing values (all columns except CouponCode)	Scanned each column for blanks/nulls	No missing values found
Data type consistency	Verified Date, Quantity, UnitPrice, and TotalPrice held consistent, correctly-formatted types throughout (no text in numeric fields, no malformed dates)	Consistent, no issues found
Formula validation	Verified $\text{Quantity} \times \text{UnitPrice} = \text{TotalPrice}$ for every row	100% match, no discrepancies
Missing values (CouponCode)	Scanned CouponCode column for blanks	309 blank entries found

4. Issue Identified and Resolved

CouponCode Column — Blank Entries

309 out of 1,200 rows (25.75%) had a blank CouponCode value. This was the only data quality issue found in the dataset.

Resolution: Blank values were treated as orders placed without a coupon, not as missing data to be removed or estimated. Each blank was filled with the label “NO COUPON” (matching the existing capitalization style of the column’s other values: FREESHIP, SAVE10, WINTER15) to make this explicit and to keep the column consistent and ready for grouping/filtering in later analysis (e.g., comparing orders with vs. without a coupon).

5. Outcome

- Dataset confirmed free of duplicate records.
- No missing values remain in any column.
- All data types are consistent across every column.
- TotalPrice values are fully validated against $\text{Quantity} \times \text{UnitPrice}$ with no discrepancies.
- CouponCode column standardized: 309 blanks replaced with “NO COUPON.”

The dataset is clean and ready for exploratory data analysis and reporting.

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